Investigate Hotel Business using Data Visualization

Project Report: Understanding Booking and Cancellation Behaviour

Labmentix Internship — Project 1

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1. Introduction and Problem Statement

Hotels earn revenue only when rooms are occupied. Understanding why guests book, when they book and why they cancel helps a hotel plan pricing, staffing, marketing and cancellation policies. Every cancelled booking is a room that could have been sold to another guest, so reducing cancellations directly protects revenue.

In this project, I acted as a data analyst for a hotel company, analysing booking records for a City Hotel and a Resort Hotel to answer three business questions:

- Which hotel type do customers book most often, and how do bookings vary by month?
- Does the length of stay affect the booking cancellation rate?
- Does lead time (the gap between booking and arrival) affect the cancellation rate?

Objective

The objective was to clean the booking data, build clear visualisations that answer each question, and turn the findings into practical recommendations for hotel management. The main deliverable is a Jupyter Notebook containing the cleaning code, charts and written analysis, supported by this report.

2. Dataset Overview

The dataset contains **119,390 booking records** and **29 columns**, covering arrivals from **2017 to 2019** (21,996 in 2017, 56,707 in 2018 and 40,687 in 2019). Each row is one booking at either the City Hotel or the Resort Hotel.

Column	Description	Used for
hotel	Hotel type: City Hotel or Resort Hotel	All three questions
is_canceled	1 if the booking was cancelled, 0 otherwise	Cancellation analysis
lead_time	Days between booking date and arrival date	Question 3
arrival_date_month / year	Month and year of arrival	Monthly trends
stays_in_weekend_nights / stays_in_weekdays_nights	Nights booked on weekends and weekdays	Length of stay (Question 2)
adults, children, babies	Number of guests	Data quality checks
adr	Average daily rate (room price per night)	Data quality checks

3. Tools and Libraries Used

Tool / Library	Purpose in this project
Python	Programming language for all data cleaning and analysis
Jupyter Notebook (Anaconda)	Interactive environment combining code, charts and written answers
Pandas	Loading the CSV, handling missing values, filtering, grouping and aggregation
NumPy	Numerical operations supporting Pandas
Matplotlib	Pie, bar and line charts; titles, axis labels and value labels
Seaborn	Chart styling and multi-series line plots by hotel type
Git and GitHub	Version control and online hosting of the project for submission

4. Data Preprocessing

Before analysis, I assessed the data for missing values, duplicates, inconsistent labels and anomalies. Each issue and the action taken is summarised below.

4.1 Missing Values

Column	Missing	Action	Reasoning
company	112,593	Filled with 0	No company recorded means a personal, non-corporate booking
agent	16,340	Filled with 0	No agent recorded means the guest booked directly
city	488	Filled with "Unknown"	Location not provided; rows kept to avoid losing booking data
children	4	Filled with 0	Most likely no children travelling

4.2 Duplicate Rows

The dataset contains **33,261 duplicate rows**. Because there is no booking ID column, identical rows are likely to be separate bookings with the same details, such as several rooms booked together for a group or family. Removing them would undercount demand, so they were **kept**.

4.3 Inconsistent Values

The *meal* column contained 1,169 entries labelled **"Undefined"**, meaning no meal package was recorded. These were recategorised as **"No Meal"** so that all bookings without a meal plan are grouped together.

4.4 Anomalies

- **Negative room price:** 1 booking had a negative *adr*, which is impossible, so it was removed.
- **Extreme room price:** 1 booking had an *adr* of 5,400, far above all other rates, and was treated as a data entry error and removed.

- **No guests:** 180 bookings recorded zero adults, children and babies. These do not represent real stays and were removed.
- **Zero room price:** about 1,959 bookings had an *adr* of 0. These were kept, as they plausibly represent complimentary stays.

Two new columns were created: **total_guests** (adults + children + babies) and **total_stay** (weekend nights + weekday nights). After cleaning, the dataset has **119,208 rows and 31 columns**.

5. Analysis and Findings

5.1 Booking Share and Monthly Trends by Hotel Type

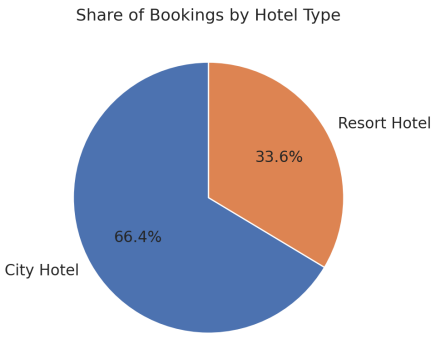


Figure 1: Share of total bookings between City Hotel and Resort Hotel.

City Hotel receives about **66.4%** of all bookings, compared with **33.6%** for the Resort Hotel. City Hotel is booked roughly **twice as often**.

When comparing months, I found that September and October appear in three years of data, while every other month appears in only two. To compare months fairly, each month's total was divided by the number of years it appears in, giving the average bookings per year.

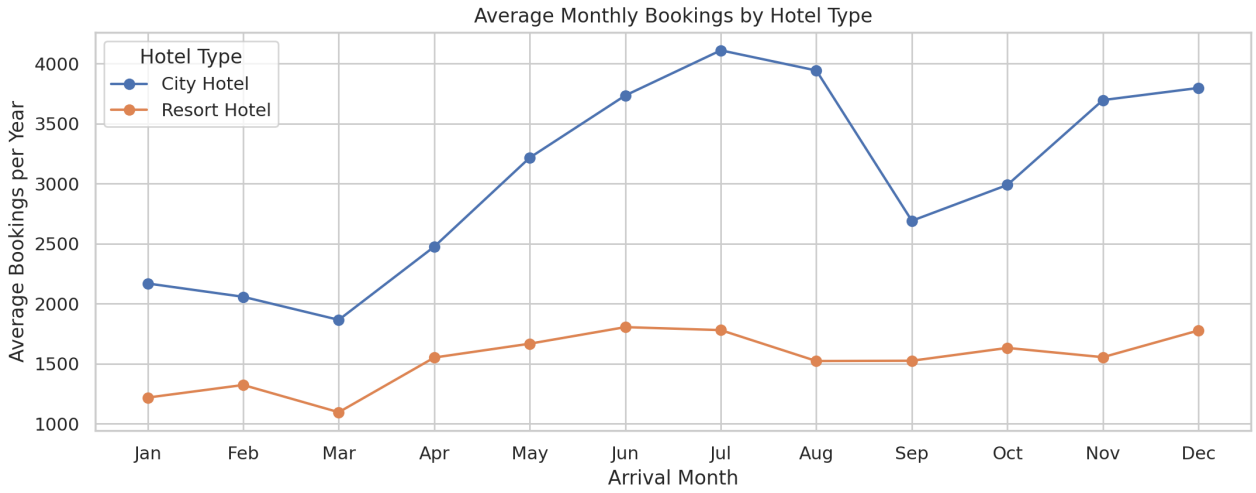


Figure 2: Average monthly bookings by hotel type, adjusted for the number of years each month appears in the data.

- **City Hotel** is busiest in **July and August** (about 4,000 bookings a month) and stays high from June to December.
- **Resort Hotel** is flatter, peaking in **June, July and December** (about 1,800 bookings a month).

- **January to March** is the quietest period for both hotels, with **March** the lowest month.

Interpretation: summer school holidays drive the mid-year peak, and Christmas and New Year raise December bookings, especially at the Resort Hotel. After the year-end holidays, travel budgets are spent and demand falls. City Hotel also serves business travellers throughout the year, which keeps its bookings consistently higher.

5.2 Impact of Stay Duration on Cancellation Rate

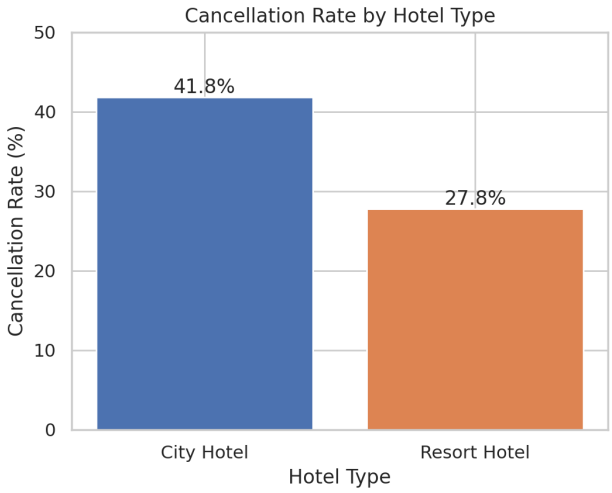


Figure 3: Percentage of bookings cancelled at each hotel type.

City Hotel has a cancellation rate of **41.8%**, compared with **27.8%** for the Resort Hotel, so City Hotel bookings are cancelled about **1.5 times as often**.

To study length of stay, stays of 1 to 7 nights were analysed individually and longer stays were grouped by week. Bookings of 0 nights (645) and over 30 nights (21) were excluded because these groups are too small or unusual to give reliable rates.

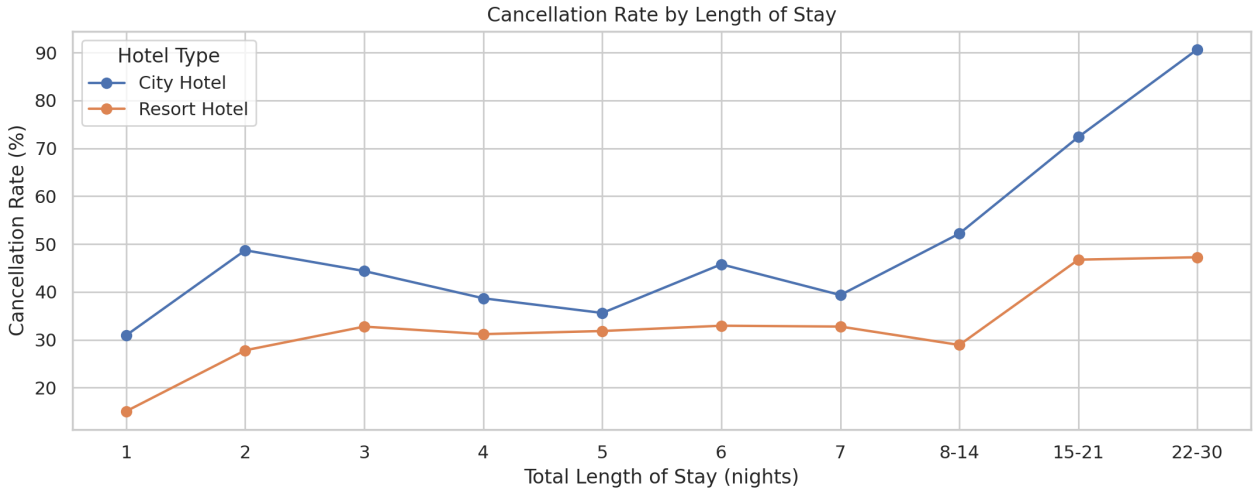


Figure 4: Cancellation rate by total length of stay, by hotel type.

- **City Hotel:** cancellations rise from **31%** for 1-night stays to 40–49% for 2–7 nights, then climb sharply to **52%** (8–14 nights), **72%** (15–21 nights) and **91%** (22–30 nights).
- **Resort Hotel:** cancellations rise from **15%** for 1-night stays to a stable 28–33% for 2–14 nights, then increase to about **47%** for stays longer than two weeks.

- Groups longer than 14 nights contain fewer than 200 bookings each, so those rates are less reliable.

Interpretation: the relationship is **positive**: longer stays are cancelled more often. It is much stronger at the City Hotel. Longer stays cost more and are often planned far in advance, leaving more time for plans to change, and some guests may reserve long stays as placeholders before confirming their plans.

5.3 Impact of Lead Time on Cancellation Rate

Lead time is the number of days between the date a booking is made and the guest's arrival date. Lead times range from 0 to 737 days, so they were grouped into 30-day intervals, with all bookings made more than a year ahead combined into one group.

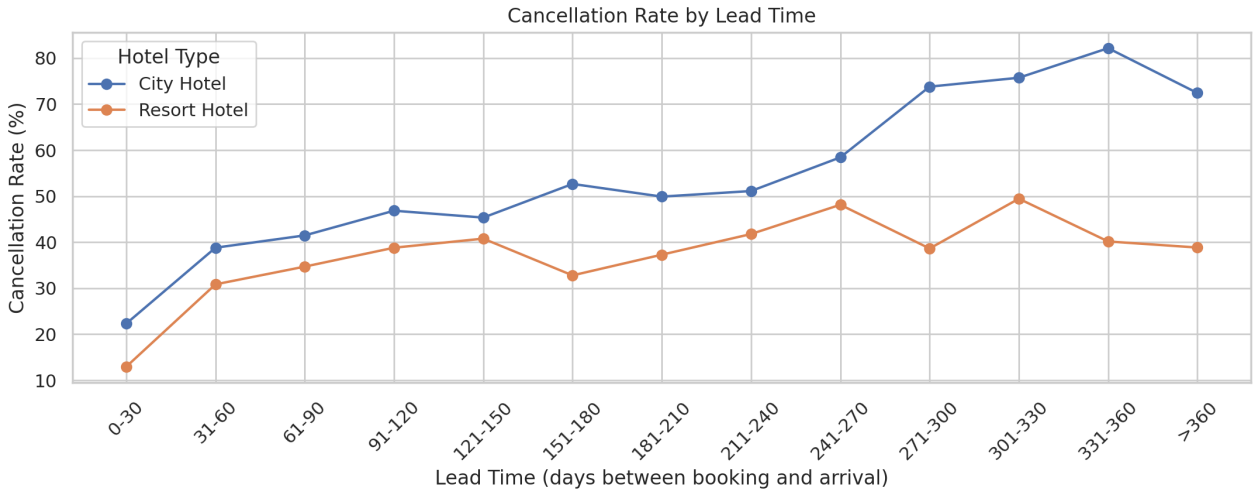


Figure 5: Cancellation rate for each 30-day lead time group, by hotel type.

- Cancellation is **lowest** for bookings made **0–30 days** before arrival: 22% at City Hotel and 13% at Resort Hotel.
- Cancellation is **highest** at **City Hotel for bookings made 331–360 days ahead (82%)**.
- **City Hotel** rises steadily with lead time, from 22% to about 50% at six months and above 70% beyond nine months.
- **Resort Hotel** is more stable: after the first month, its rate stays between about **31% and 50%**.

Interpretation: resort stays are usually planned holidays, so guests who book early tend to be committed. City Hotel serves more business and event travellers, whose plans change often, and city guests have many nearby alternatives, so early bookers have more time to find a cheaper option and cancel. Bookings made within a month are usually for trips that are already certain.

6. Summary of Key Findings

Business question	Key finding
Which hotel type is booked most?	City Hotel receives 66.4% of bookings, about twice the Resort Hotel. Demand peaks in summer and December and is lowest in January–March.
Does length of stay affect cancellation?	Yes, positively. Longer stays are cancelled more often, rising to over 70% at City Hotel for stays longer than two weeks. City Hotel's overall rate (41.8%) is higher than Resort Hotel's (27.8%).
Does lead time affect cancellation?	Yes. Bookings made within 30 days are cancelled least. City Hotel cancellations climb to 82% for bookings made 331–360 days ahead, while Resort Hotel stays within 31–50%.

7. Recommendations

7.1 Hotel Type and Seasonality

- **Grow the Resort Hotel in its low season:** offer winter getaway packages (discounted multi-night stays with meals or activities) for January–March, promoted in November–December.
- **Dynamic pricing in peak months:** raise rates in June–August and December, when demand is highest.
- **Plan operations around the cycle:** hire seasonal staff before peaks; schedule renovations and staff leave for January–March.

7.2 Stay Duration and Cancellations

- **Partial non-refundable deposit for long stays:** for example, the first night's charge for bookings of 8 nights or more, where cancellation reaches 52–91% at City Hotel.
- **Discounted non-refundable rates:** a lower price in exchange for no cancellation locks in revenue.
- **Allow changes instead of cancellations:** let guests shorten a stay or move dates free of charge, so the hotel keeps part of the revenue.

7.3 Lead Time and Cancellations

- **Deposits for far-ahead bookings:** require a deposit for bookings made more than six months in advance, refundable only up to a set date before arrival.
- **Reminders before arrival:** confirm plans with guests 60 and 30 days before arrival, so cancelled rooms can be resold in time.
- **Free rescheduling:** guests whose plans change are more likely to move dates than cancel.
- **Early-booking discounts tied to upfront payment:** reward early planners without encouraging placeholder bookings.

7.4 Recommendation with the Biggest Impact

Requiring deposits for City Hotel bookings made more than six months in advance would likely have the biggest impact. City Hotel receives two-thirds of all bookings (Figure 1) and has the higher cancellation rate (Figure 3), so most cancelled bookings occur there. For bookings made more than six months ahead, its cancellation rate ranges from about 50% to 82% (Figure 5). The policy targets

only these high-risk bookings, leaving last-minute bookings, which rarely cancel, unaffected.

8. Challenges and Solutions

Challenge	Solution implemented
Large share of missing values in *company* (94% of rows) and *agent*	Filled with 0 rather than dropping rows, since a missing value carries meaning (no company or no agent involved). This kept all bookings in the analysis.
33,261 duplicate rows with no booking ID to confirm whether they were errors	Kept the rows after reasoning that identical bookings are likely group or family bookings; documented the decision so it can be reviewed.
Unclear "Undefined" category in the *meal* column	Merged it with "No Meal", since both indicate no meal package.
Invalid records: negative and extreme room prices, and bookings with zero guests	Removed the 182 affected rows after confirming they could not represent real stays; kept zero-price bookings as plausible complimentary stays.
Months appear in different numbers of years (September and October in three years, others in two), which inflated their raw totals	Calculated average bookings per year for each month, so all months are compared on the same basis.
Very few bookings at long stay lengths and long lead times, giving unstable rates	Grouped stays longer than 7 nights by week and lead time into 30-day intervals; excluded very small groups and noted where sample sizes are low.
Chart code failed with *AttributeError: 'Rectangle' object has no property 'legend'* because the installed Seaborn version was older	Rewrote the bar chart using Matplotlib directly, which works across library versions.
Version control was not set up while the project was in progress	Initialised a Git repository after completion, organised in a single Labmentix folder for this and future projects, and published it to GitHub.

9. Conclusion

This project showed that City Hotel is the more popular choice but also the riskier one: it receives two-thirds of bookings, yet loses a far larger share of them to cancellations, especially for long stays and bookings made many months in advance. The Resort Hotel has steadier cancellation behaviour but weaker demand early in the year.

The recommendations focus on protecting revenue where cancellation risk is highest, through deposits, non-refundable rates and flexible rescheduling, and on filling rooms in the low season through targeted packages and seasonal pricing. Together, these actions can help the hotel improve occupancy and reduce revenue lost to cancellations.